Comprehensive Screening of Microplastic Measurement Methods in Surface Water

Established Laboratory Approaches, Electrical Impedance, and Acoustic Technologies

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Abstract

This working review re-examines **10,833 preliminary include-candidate papers** from the combined Web of Science–Scopus search, including **6,654 records** initially tagged as laboratory methods. The original 14-category Laboratory Methods screening is presented in full alongside a targeted re-screen for electrical impedance, acoustics, microfluidics, flow cytometry, and advanced imaging. The two screens use different candidate sets and text fields; their counts are shown separately and must not be added together. The emphasis is on *working principle, what is actually measured, and a testable validation pathway* for a portable instrument estimating microplastic (MP) concentration in river and lake surface water. Counts represent title/abstract/keyword mentions, not proven sensor performance or completion of systematic-review eligibility.

1 Search scope and additional methods identified

The merged search produced 18,942 records; 7,318 duplicates and 727 document-type records were excluded before the 10,897-record title/abstract screening set. Of these, 10,833 were provisionally included and 64 need manual assessment. The first screen comprises 14 broad method families among 6,654 Laboratory Methods candidates (Fig. 1 and Table 1). The follow-up, targeted technology screen searches titles, abstracts and keywords across the full 10,833 candidate set (Table 2 and Fig. 2). These denominators and field scopes differ; frequencies are descriptive and cannot be compared as prevalence rates or combined by addition. *A paper may mention a method without using it*; ultrasound terms may describe sonication during preparation or water treatment, while acoustic focusing typically performs separation rather than direct particle counting. The older Laboratory Methods tag should not be treated as a definitive filter.

1.1 Original broad methodology screening: 14 categories

The initial Laboratory Methods classification grouped the **6,654 method-tagged candidate records** into 14 overlapping categories. Figure 1 reproduces the original broad-category frequency bar chart, with Table 1 listing the exact counts and within-subset percentages. These are automated keyword matches, not independently verified experimental protocols. In particular, an article labelled “FTIR” may discuss the technique in a review rather than report its own FTIR measurements.

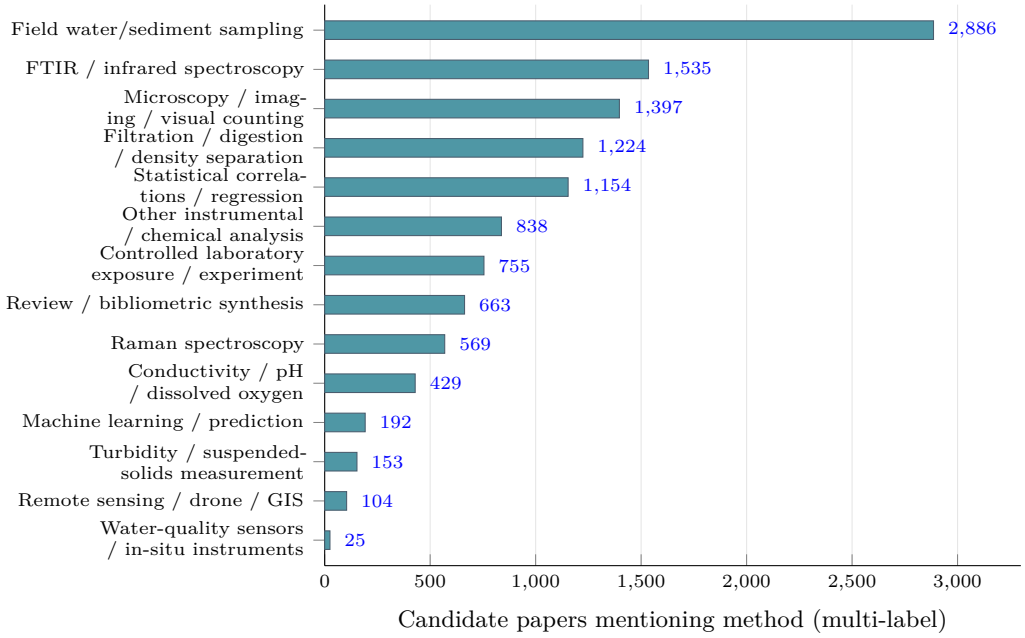


Figure 1: Original 14-category Laboratory Methods screening: keyword-tagged papers within 6,654 Laboratory Methods candidates. Papers can match multiple categories; these bars are neither counts of instrument deployment nor measures of performance.

Table 1: Full broad-method classification and share of the original 6,654 Laboratory Methods candidates (multiple assignments permitted).

Original method category	Papers	Within subset
Field water/sediment sampling	2,886	43.4%
FTIR / infrared spectroscopy	1,535	23.1%
Microscopy / imaging / visual counting	1,397	21.0%
Filtration / digestion / density separation	1,224	18.4%
Statistical correlations / regression	1,154	17.3%
Other instrumental / chemical analysis	838	12.6%
Controlled laboratory exposure / experiment	755	11.3%
Review / bibliometric synthesis	663	10.0%
Raman spectroscopy	569	8.6%
Conductivity / pH / dissolved oxygen	429	6.4%
Machine learning / prediction	192	2.9%
Turbidity / suspended-solids measurement	153	2.3%
Remote sensing / drone / GIS	104	1.6%
Water-quality sensors / in-situ instruments	25	0.4%

1.2 Targeted re-screen of underrepresented detection technologies

The complementary re-screen covers the **10,833 preliminary include candidates**, not only the Laboratory Methods subset. Table 2, Figure 2, and Figure 3 identify promising research directions that may be missed by broad categories. For example, “ultrasonication” can refer to preparing a sample rather than detecting or quantifying MPs. The technology-specific counts must therefore be validated through paper-level screening.

Table 2: Re-screened technology mentions, drawn from the same 10,833 candidate publications. The last column indicates previously tagged Laboratory Methods candidates; categories overlap.

Technique/keyword family	All candidates	Lab-tagged subset
Electrical impedance / EIS	5	5
Ultrasound / ultrasonication	42	31
Acoustic / photoacoustic	13	10
Electrochemical / voltammetric	55	48
Microfluidics	22	18
Flow cytometry	35	34
Digital holography	13	12
Hyperspectral imaging	31	30
Polarisation-sensitive scattering	3	3
Photoacoustic (also in acoustic)	3	3
Nile Red staining	138	127
Pyrolysis / Py–GC–MS	191	168
FTIR / infrared	1,909	1,882
Raman spectroscopy	664	664

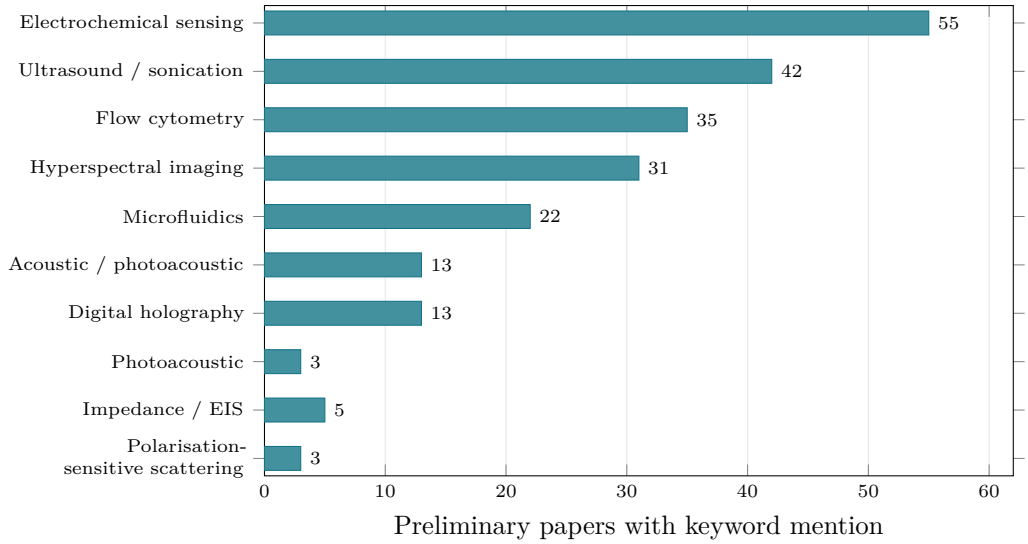


Figure 2: New targeted technology screen among the 10,833 candidates (overlapping keyword matches). Acoustic/ultrasonic counts include non-detection uses.

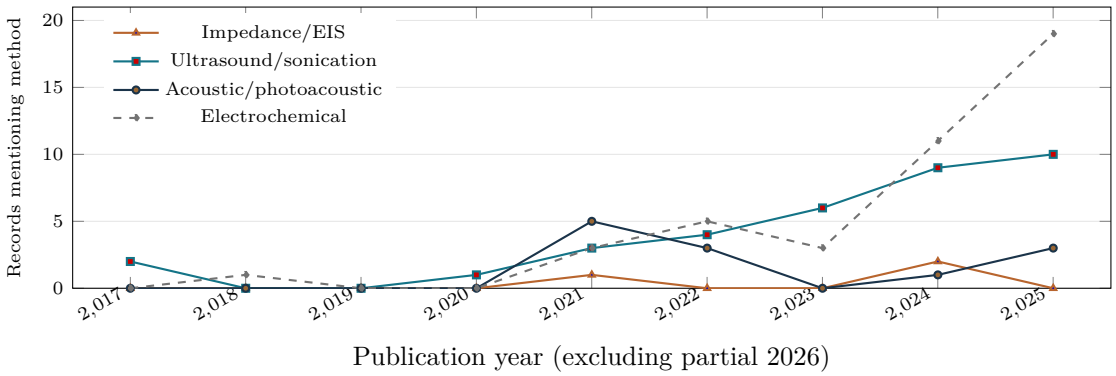


Figure 3: Annual *keyword mentions* in screened candidates. Increasing use of a term is not evidence of sensor sensitivity, field readiness, or direct MP detection. Sonication and acoustic terms include non-detection applications.

2 Priority A: Electrical impedance and electrochemical sensing

Electrical impedance spectroscopy (EIS) measures the complex ratio between a small AC excitation and electrical current, $Z(f) = V(f)/I(f)$. An MP-related event might alter current distribution or an electrode interface, but the measured response also depends strongly on dissolved ions, temperature, geometry, electrode polarisation and fouling. It therefore cannot be equated directly with particles/L. *Two distinct architectures* must be evaluated: (i) bulk or electrode-surface spectra as a function of concentration, and (ii) a constrained microfluidic path where individual particles perturb an electrical field. Blevins *et al.* discuss laboratory prototypes including impedance spectroscopy and dielectrophoresis, while emphasising the trade-offs in field deployability and information output [1].

Kim *et al.* report a *material-specific electrode* using porous $NiMoO_4$ nanotubes for PVC microplastics; cyclic voltammetry and EIS were used, with an abstract-reported detection limit near **18 ppm** and $R^2 \simeq 0.978$ [2]. This is a promising *PVC-focused laboratory calibration*, **not** a general river-water particle counter. The original screening also includes microfluidic electrical sensing using a Wheatstone-bridge/MXene electrode architecture [3], and a microfluidic *microwave* resonator detecting permittivity-induced frequency shifts from polyethylene microspheres [4]. Microwave permittivity sensing is not the same as low-frequency EIS.

Table 3: Working-principle distinctions needed before choosing an electrical prototype. These are engineering interpretations informed by screened papers.

Approach	Measured signal and potential MP information	Main difficulty / proposed verification
Bulk impedance/EIS [1]	Frequency-dependent current/phase affected by sample impedance.	Separate MP response from salinity, pH, temperature and electrode artefacts.
Functionalised electrochemical electrode [2]	Change in voltammetric/electrode-interface signal after particle–surface interaction.	Polymer specificity, electrode ageing, detection limit in actual river matrix.
Electro-microfluidic sensing [3]	Particle perturbation in a confined electrode channel; electrical bridge output.	Flow rate, clogging, particle size, ionic-strength dependence, events/L.
Microfluidic microwave resonator [4]	Shift in resonant frequency from dielectric loading / permittivity change.	Small shifts; temperature and water permittivity can dominate.

3 Priority B: Ultrasound, acoustophoresis and photoacoustics

Ultrasonic detection sends sound through water and measures changes in attenuation, scattering or acoustic arrival. These signals depend on particle size, acoustic contrast, suspended mineral matter and bubbles; non-plastic debris can mimic a plastic signal. This must be distinguished from **sonication used for preparation**, which is a processing step, not a concentration sensor [1].

Acoustophoresis uses a standing acoustic wave to drive suspended particles toward pressure nodes or antinodes according to their density/compressibility contrast and size. It is principally an *enrichment/separation* technique: a downstream microscope or flow sensor is still needed to report polymer-confirmed MP abundance [1], [5]. In model particles and laundry wastewater, Ohori *et al.* report **100 mL/min throughput**, over **1,000-fold enrichment**, and over **90% recovery** for tested 13–100 μm sizes. Such figures should not be interpreted as a field microplastic-concentration detection limit [5].

Photoacoustics uses modulated light absorption to generate an acoustic response. Grzegorczyk *et al.* compared it with diffuse reflectance for model plastic and algal suspensions; their abstract reports greater plastic-versus-biota contrast from the reflectance method in the tested system [6]. It is separate from conventional ultrasound scattering and currently warrants bench-level feasibility tests rather than claiming direct river deployment.

Table 4: Representative acoustic evidence and its implications for a surface-water prototype.

Mode	Source-supported observation	Role in project / outstanding issue
Ultrasound-based physical characterisation [1]	Prototype sensing route discussed alongside impedance and optical instruments.	Test selectivity versus silt, bubbles and microalgae.
Acoustic focusing [5]	Continuous particle recovery and enrichment demonstrated in spiked and laundry samples.	Add downstream particle counting; determine recovery and total sampled volume.
Photoacoustic / reflectance [6]	Model mixtures gave method-dependent spectral responses; reflectance gave higher contrast in the abstract.	Compare plastic–algae discrimination with conventional optical measurements.

4 Other overlooked methods worth a focused pilot

Table 5: Selected detection or sample-processing approaches retrieved from the original candidate collection. A citation demonstrates the studied approach, not universal field performance.

Technique	Principle / relevant published demonstration	Suitability or limitation
Flow cytometry [7], [8]	Hydrodynamically aligned particles pass laser scatter / fluorescence sensors; classifier separates particles from background.	Potential high-throughput counts; blanks, matrix classification and polymer validation essential.
Digital holography [9], [10], [11]	Interference patterns encode particle shape/phase; polarimetric variants improve contrast; computational reconstruction/classification.	Strong in-situ potential; model accuracy and concentration accuracy require independent testing.
Hyperspectral imaging [12], [13]	Polymer-related wavelength signatures from VIS/NIR/SWIR images; classifier assigns material type.	Useful for collected particles or sufficiently large visible MPs; water absorption and turbid conditions matter.
Polarised light scattering [14]	Polarisation dependence of scattered light carries particle/material features.	Laboratory proof-of-principle; natural-water discrimination and flow calibration required.
Microfluidic pretreatment/counting [15]	Integrates particle digestion/filtration/counting in a small channel/device.	Sample throughput, clogging and blank control must be quantified.
Nile Red and microscopy [7], [16]	Staining enhances fluorescence of suspected plastic particles.	Fast screening but potential false positives; FTIR/Raman confirmation advised.
Polymer-specific reference [17], [18]	Filtration/spectral particle identification and complementary thermal mass measurement.	Necessary laboratory ground truth; mass/L and particles/L are not equivalent.

5 Proposed chemistry laboratory and field validation

The project requires an independently measured reference *particle concentration* (for example, polymer-confirmed particles/L above a stated size cutoff). A chemistry lab can filter a recorded sample volume, apply recovery and contamination controls, count particles and confirm polymer identity using micro-FTIR or Raman; Py–GC–MS measures complementary polymer *mass* and should not be substituted for a particle count [17], [18]. For tests involving acoustic enrichment, count both the concentrated and bypass streams and report recovery, not just concentration enhancement.

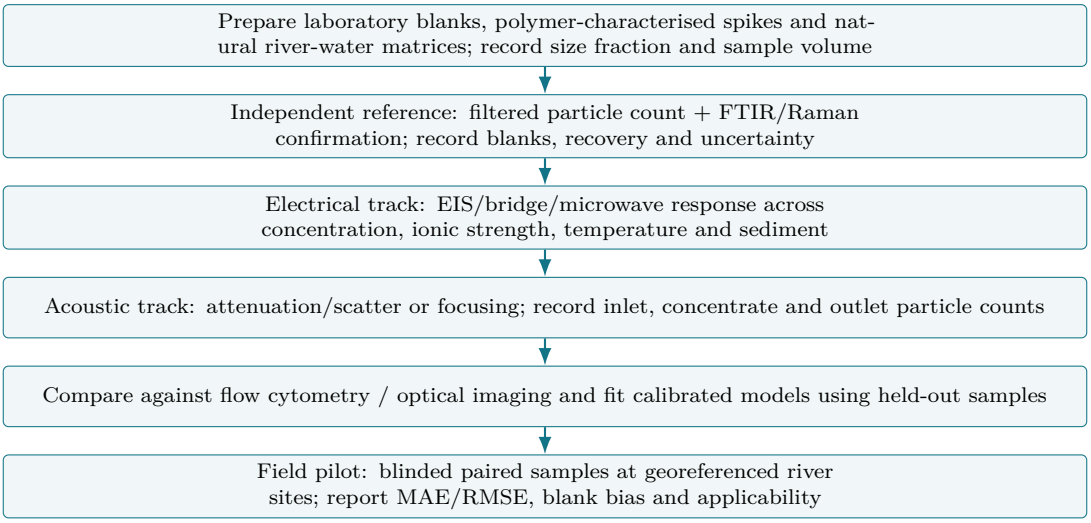


Figure 4: Proposed validation path, not an already demonstrated integrated instrument.

Table 6: Proposed pilot matrix to discuss with the chemistry laboratory. Levels are illustrative design factors; choose numeric concentrations after confirming feasible instrumentation and reference standards.

Controlled variable	Suggested test design	Decision criterion
Blanks	Filtered-water, reagent, procedural, and ambient-air blanks.	Baseline noise and contamination-corrected counts.
MP spikes	At least blank + low/medium/high known spike levels; replicates.	Sensitivity, detection limit, linearity and bias to lab reference.
Particle diversity	PE, PS and PET; size classes and fibers/fragments if available.	Material/size dependence and false-negative rates.
Conductivity / temperature	Fixed MP count, varying dissolved-salt level and temperature.	Test whether EIS detects ions rather than MP particles.
Nonplastic background	Sediment, microalgae and air bubbles, separately and combined.	Specificity; false-positive rate for electrical/acoustic/optical systems.
Acoustic recovery	Collect inlet, bypass and concentrated fractions at logged flow.	MP mass/number balance and recovery versus size.
River validation	Withheld sites/dates with paired chemistry-lab tests.	Prediction error, transferability and map uncertainty.

For a validated particle-count reference use

$$C_{\mathrm{ref}} = \frac{N_{\mathrm{confirmed}} - N_{\mathrm{blank}}}{V_{\mathrm{sample}}\, r} \qquad (\text{particles/L}), \tag{1}$$

where r is a measured recovery fraction (if an experimentally justified correction is used). Report uncorrected counts and recovery uncertainty if r is not known. For an acoustic concentrator, an additional sample-specific recovery is necessary; a high enrichment factor alone does not establish accurate source-water concentration.

6 Conclusion and near-term engineering decision

First priority: develop a reproducible particle-count reference with the chemistry laboratory. **Parallel prototypes:** test EIS/electro-microfluidic sensing and acoustic concentration coupled to downstream counting. **Comparison benchmarks:** flow cytometry, polarised scattering or holographic imaging are suitable for targeted feasibility comparisons. A method progresses to river mapping only if it measures (or predicts with validated uncertainty) polymer-confirmed MPs across particle types and representative sediment, algae, salt and temperature conditions. This is a *proposal from the candidate-literature evidence map*; definitive method-performance comparisons await full-text appraisal and the pilot results.

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